1. Mnożenie wielomianów 3

Wyznacz wielomiany f(x)= u(x)* w(x) oraz g(x)=2w(x)-$[u(x)]^2$.

a) u(x)=x^2-5x, w(x)=x^3+x^2+2,

b) u(x)=$-2x^3+2x^2-x+1$, w(x)=$4x^2-x+2$

a) $u(x) = x^2 - 5x$

$w(x) = x^3 + x^2 + 2$

① Wyznaczenie $f(x) = u(x) \cdot w(x)$

$f(x) = (x^2 - 5x)(x^3 + x^2 + 2) = x^5 + x^4 + 2x^2 -$

$- 5x^4 - 5x^3 - 10x = x^5 - 4x^4 - 5x^3 + 2x^2 - 10x$

② Wyznaczenie $g(x) = 2 \cdot w(x) - [u(x)]^2$

$(a-b)^2 = a^2 - 2ab + b^2$

$g(x) = 2 \cdot (x^3 + x^2 + 2) - (x^2 - 5x)^2 =$

$= 2x^3 + 2x^2 + 4 - (x^4 - 10x^3 + 25x^2) =$

$= 2x^3 + 2x^2 + 4 - x^4 + 10x^3 - 25x^2 =$

$= -x^4 + 12x^3 - 23x^2 + 4$

Odp: $f(x) = x^5 - 4x^4 - 5x^3 + 2x^2 - 10x$,

$g(x) = -x^4 + 12x^3 - 23x^2 + 4$.

b) $u(x) = -2x^3 + 2x^2 - x + 1$

$w(x) = 4x^2 - x + 2$

① Wyznaczenie $f(x) = u(x) \cdot w(x)$

$f(x) = (-2x^3 + 2x^2 - x + 1)(4x^2 - x + 2) =$

$= -8x^5 + 2x^4 - 4x^3 + 8x^4 - 2x^3 + 4x^2 - 4x^3 + x^2 -$

$- 2x + 4x^2 - x + 2 =$

$= -8x^5 + 10x^4 - 10x^3 + 9x^2 - 3x + 2$

② Wyznaczenie $g(x) = 2 \cdot w(x) - [u(x)]^2$

$g(x) = 2(4x^2 - x + 2) - (-2x^3 + 2x^2 - x + 1)^2 =$

$= 8x^2 - 2x + 4 - (-2x^3 + 2x^2 - x + 1)(-2x^3 + 2x^2 - x + 1) =$

$= 8x^2 - 2x + 4 - (4x^6 - 4x^5 + 2x^4 - 2x^3 - 4x^5 + 4x^4 -$

$- 2x^3 + 2x^2 + 2x^4 - 2x^3 + x^2 - x - 2x^3 + 2x^2 - x + 1) =$

$= 8x^2 - 2x + 4 - (4x^6 - 8x^5 + 8x^4 - 8x^3 + 5x^2 - 2x + 1) =$

$= 8x^2 - 2x + 4 - 4x^6 + 8x^5 - 8x^4 + 8x^3 - 5x^2 + 2x - 1 =$

$= -4x^6 + 8x^5 - 8x^4 + 8x^3 + 3x^2 + 3$

Odp: $f(x) = -8x^5 + 10x^4 - 10x^3 + 9x^2 - 3x + 2$,

$g(x) = -4x^6 + 8x^5 - 8x^4 + 8x^3 + 3x^2 + 3$